Adil Shamim

AI Engineer

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I think from first principles and build products from zero to one.
I turn customer problems into practical AI systems, from architecture to team delivery.

Experience

CTO & AI Engineer | FinAI (UK, Remote) Jan 2026 – Present

AI Intern → AI Engineer → CTO

- Lead the engineering team, assign development tasks, review outcomes and guide delivery against product priorities.
- Design the product and system architecture for finance AI workflows, including backend services, retrieval, tool integration and evaluation.
- Work directly with customers to understand their workflows, turn requirements into development tasks and refine the product through feedback.

Founder & AI Engineer | ReWoo (rewoo.tech) Jul 2026 – Present

- Develop AI employees that help one founder run business functions, using defined responsibilities, persistent memory, and workflow context.
- Automate routine work within each agent's role, with quality checks before customer-facing output is released.

Research Intern | ELITE Research Lab LLC (Queens, NY — Remote) Expected end: Dec 2026

- Investigate AI hallucination detection and contribute to research papers and a journal manuscript in preparation.

Founder | Toolly (open-sourced) Jun 2025 – Mar 2026

- Built an AI discovery platform attracting approximately 4,000–5,000 monthly visitors; automated tool submissions and moderation.

Selected Engineering Projects

- **Agentic AI Eval Harness | Open Source**
  Built a CI gate to catch agent regressions before release by evaluating complete runs against 280 generated, hash-pinned cases and blocking merges when results fall below configured thresholds. Separated agent, evaluator, and infrastructure failures to help developers identify what broke; used ablation runs to test retry and tool-validation behavior. Reported measurements use a deterministic backend.
- **Agentic RAG Platform | Reference Implementation**
  Built a reference system for source-backed answers over changing organizational documents, combining pgvector and PostgreSQL full-text search with reciprocal rank fusion and BGE reranking. Applied permissions before retrieval, validated citations against evidence, and traced runs with OpenTelemetry/Langfuse; added evaluation gates to check changes before release.

Technical Skills

Programming & Backend Python, SQL, TypeScript, Bash, FastAPI, PostgreSQL
AI Systems LangGraph, tool calling, RAG, hybrid retrieval, reranking, agent evaluation
Testing & Integration Scikit-learn, Pytest, API integration, access controls, CI regression gates
ML & Speech PyTorch, Hugging Face Transformers, Whisper, pyannote.audio, WeSpeaker
Delivery & Observability Docker, AWS, GitHub Actions CI/CD, OpenTelemetry, Langfuse

Research & Recognition

- **First author** — *Bengali Hallucination Detection with Retrieval and Programs: A System Study* — inference-only judging architecture: response-conditioned query rewriting, six hybrid retrieval routes, and a router dispatching each case to a 27B textual judge or a 14B agent that reasons directly or executes generated Python. 0.956 private macro-F1 on the Olikbochon challenge.
- **First author** — *Bangla Diarizz: Domain-Adapted Speaker Diarization for Bengali Long-Form Audio*, BUET CSE Fest 2026 / DL Sprint 4.0 (107 teams): fine-tuned pyannote segmentation plus WeSpeaker ResNet34-LM embeddings with gap-stitching post-processing; 56% wall-clock speedup (1h22m → 36m), private DER 0.286, rank 19.
- **ICISET 2026 — Accepted for oral presentation:** *Automated End-to-End Signal Processing and Interpretable Machine Learning for Superconducting Qubits*. Conference acceptance rate: 27%.
- **First Runner-Up — Legal Tech Hackathon 2026**, Bangladesh.
- **Kaggle Notebooks Master** (167 / 62,365), top 1% finish (29 / 4,082) | startup funding, ITBI Pitch Fest 2026 | 920+ GitHub stars.

Education & Certifications

Computer Science & Engineering | BNIST Feb 2023 – Present

Certifications AI Engineer for Developers Associate, DataCamp (2026) | Machine Learning, Stanford / Coursera | CS50, Harvard